
Institute of Quantum Computing, University of Waterloo, Canada
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[website](#)

RESEARCH INTERESTS

Quantum Key Distribution, Quantum and Classical Cryptography, Open Quantum Systems.
5 peer-reviewed publications and 7 pre-prints.

ACADEMIC BACKGROUND

Ph.D. Physics (Quantum Information), GPA : 92/100 2023-Present
[Institute for Quantum Computing, University of Waterloo](#)
Mike and Ophelia Lazaridis Fellowship.

- Research on security analysis of practical quantum key distribution protocols.
- Relevant Coursework (Quantum Information and Cryptography): Quantum Error-correction and Fault-Tolerance, Quantum Information Processing, Implementation of Quantum Information Processing, Theory of Quantum Information, Applied Quantum Cryptography, Applied Cryptography, Mathematics of Public Key Cryptography, Privacy Data and Security.
- Contributor to [Open QKD Security](#) software package for numerical QKD analysis.

M.Sc. Physics (Quantum Information), GPA : 96/100 2020-2022
[Institute for Quantum Computing, University of Waterloo](#)

B.Sc. (Research) Physics, GPA : 9.2/10 2016-2020
[Indian Institute of Science](#)

- Graduated First Class with Distinction.
- Relevant Coursework (Quantum Information and Cryptography) : Quantum Mechanics 1, Quantum Mechanics 2, Quantum Computation, Quantum Measurements and Control, Quantum Information Theory, Information Theory, Theoretical Foundations of Cryptography.

PUBLICATIONS

- [P1] ***Phase error estimation for passive detection setups with imperfections and memory effects.***
Zhiyao Wang, Devashish Tupkary, Shlok Nahar, [arXiv link](#).
- [P2] ***Imperfect detectors for adversarial tasks with applications to quantum key distribution.***

Shlok Nahar, Devashish Tupkary, Norbert Lütkenhaus, [arXiv link](#). Under review at Quantum.

- [P3] ***Security of quantum key distribution with source and detector imperfections through phase-error estimation.***
Guillermo Currás-Lorenzo, Margarida Pereira, Shlok Nahar, Devashish Tupkary, [arXiv link](#).
- [P4] ***QKD security proofs for decoy-state BB84: protocol variations, proof techniques, gaps and limitations.***
Devashish Tupkary, Ernest Y-Z Tan, Shlok Nahar, Lars Kamin, Norbert Lütkenhaus, [arXiv link](#).
- [P5] ***A consolidated and accessible security proof for finite-size decoy-state quantum key distribution.***
Jerome Wiesemann, Jan Krause, Devashish Tupkary, Davide Rusca, Nino Walenta, Norbert Lütkenhaus, [arXiv link](#). Under review at Quantum.
- [P6] ***Improved finite-size effects in QKD protocols with applications to decoy-state QKD.***
Lars Kamin, Devashish Tupkary, Norbert Lütkenhaus, [arXiv link](#). Under review at Physical Review Research. Combined with other work, accepted talk at [QCRYPT 2024](#) and shortlisted for best student theory submission.
- [P7] ***Phase error rate estimation in QKD with imperfect detectors.***
Devashish Tupkary, Shlok Nahar, Pulkrit Sinha, Norbert Lütkenhaus, [arxiv link](#). Under review at Quantum. Combined with other work, accepted talk at [QCRYPT 2024](#) and shortlisted for best student theory submission.
- [P8] ***Postselection technique for optical prepare-and-measure QKD protocols.***
Shlok Nahar, Devashish Tupkary, Yuming Zhao, Norbert Lütkenhaus, Ernest Y.-Z. Tan, [PRX Quantum 5, 040315](#). Combined with other work, accepted talk at [QCRYPT 2024](#) and shortlisted for best student theory submission.
- [P9] ***Security Proof for Variable-Length Quantum Key Distribution.***
Devashish Tupkary, Ernest Y.-Z. Tan, Norbert Lütkenhaus, [Phys. Rev. Research 6, 023002](#). Combined with other work, accepted talk at [QCRYPT 2024](#) and shortlisted for best student theory submission. Awarded the Cryptoworks21 Distinguished Student Paper Award.
- [P10] ***Using Cascade in Quantum Key Distribution.***
Devashish Tupkary, Norbert Lütkenhaus, [Phys. Rev. Applied 20, 064040](#).
- [P11] ***Searching for Lindbladians obeying local conservation laws and showing thermalization.***
Devashish Tupkary, Abhishek Dhar, Manas Kulkarni, and Archak Purkayastha, [Phys. Rev. A 107, 062216](#). Invited Talk at [CQIQC Seminar](#), University of Toronto, 10th Feb 2023. [video link](#).
- [P12] ***Fundamental Limitations in Lindblad Descriptions of systems weakly coupled to baths.***

SCHOLARSHIPS AND AWARDS

- [Cryptoworks21 Distinguished Student Paper Award](#) - For [PP9]. 2025
- [Mike and Ophelia Lazaridis Fellowship](#) - Fellowship during PhD, includes tuition and a yearly stipend of 36000 CAD. 2023-Present
- [Long Term Visiting Student Program, International Centre for Theoretical Studies - Tata Institute of Fundamental Research](#) - Among a dozen students selected for a year-long funded research project. 2019
- [Madhava Mathematical Competition](#). *Rank 6* - A nationwide mathematics competition for undergraduate students. Attended the Madhava Mathematics Competition camp at the [Indian Statistical Institute, Bangalore](#). 2017
- [Kishore Vaigyanik Protsahan Yojana \(KVPY-SA Fellowship\)](#). *Rank 39* - Fellowship during undergraduate degree, includes a yearly stipend of 80000 INR. 2016-2020
- [National Bal Shree Honour for Creative Scientific Innovation](#) - One of the three highest national honours for children by the Government of India. 2016
- [IIT-JEE \(Advanced\)](#). *Rank 312* - A nationwide entrance exam for college admissions to the Indian Institute of Technology (IITs). 2016
- [Indian National Mathematics Olympiad](#) - *Ranked among the top 70 nationwide* 2016
- [Qualified Indian National Physics Olympiad](#) - *Ranked among the top 35 nationwide*. 2016
- [National Talent Search Examination](#) - *Among 1000 students awarded the NTSE Scholarship nationally in each year.* 2012-2016

INVITED AND CONTRIBUTED TALKS

- Contributed Talk at [TQC 2025](#) on “Phase error rate estimation in QKD with imperfect detectors.” Based on [PP7]. 16 Sept 2025
- Invited Tutorial Talk at [QCrypt 2025](#) on “Security Proofs for Decoy-State BB84 and Their Performance.” Based on [PP4]. 26 August 2025
- Lecture at [Cryptoworks21](#), University of Waterloo, on “An Introduction to Quantum Key Distribution.” 22 July 2025
- Contributed Talk at [QCrypt 2024](#) on “Security Proof for Variable-Length Quantum Key Distribution” and “Variable-Length QKD Security Proof for Imperfect Detectors through Phase-Error Estimation.” Based on [PP9],[PP7]. 2 September 2024
- [Invited Talk](#) at [CQIQC Seminar](#), University of Toronto, on “Lindbladians Obeying Local Conservation Laws and Showing Thermalization”. Based on [PP11][PP12]. 10 February 2023

SEMINARS

- Talk at [IISc Bengaluru](#), [IISER Pune](#) on “Extending Quantum Key Distribution security proofs to practical detectors”. Based on [PP7][PP1][PP3]. 11th, 12th Sept 2025
- Talk at [IIT Bombay](#) on “An Introduction to Quantum Key Distribution”. 1st Sept 2025
- Talk at [ICTS-TIFR](#) on “ A brief introduction to Quantum Key Distribution” 4th Jan 2024
- Contributed Talk at QKD Security Proofs workshop on “Phase error estimation with basis-efficiency mismatch.” Based on [PP7] 31st Sept 2024
- Contributed Talk at QKD Security Proof workshop on “Postselection technique” and “Variable-length QKD.” Based on [PP9][PP8] 12th, 13th Sept 2023
- Contributed Talk at QKD Security Proof workshop on “Improved processing of classical data in QKD” 21st Sept 2022

SUPERVISION

- Supervised Zhiyao Wang, research project at IQC–Waterloo, leading to [PP1]. 2024–2025
- Supervised Jerome Wiesemann, research project leading to [PP5]. 2024–2025
- Supervised Soumyadeep Sarma, Bachelor’s thesis at ICTS–TIFR, Bengaluru. 2024–Present

POSITIONS OF RESPONSIBILITY

- Co-Organizer of the QKD Security Proof Workshop at IQC Waterloo. 2022, 2023, 2024
- Co-Organizer of the weekly IQC Student Seminar Series. 2023–2024